Lesson 1.3: Backend engineering and databases

What you learn

- **Node.js and Express.** Node runs JavaScript on the server. Express adds routes and middleware: functions that run in order on each request (logging, auth, validation) before the route handler.
- **ORMs: Prisma and Drizzle.** An ORM maps database tables to code. Prisma uses a schema file and generates a typed client; Drizzle writes the schema in TypeScript and stays close to SQL. Both give migrations.
- **PostgreSQL.** A relational database: tables, keys, joins, constraints and transactions. Choose it when data has clear relationships and integrity matters.
- **MongoDB.** A document database that stores JSON-like documents. Choose it when records vary in shape; design around how the data is read.

Tools and resources (official sites, not checked live)

- Node.js
- Express
- Prisma
- Drizzle ORM
- PostgreSQL
- MongoDB

ASSIGNMENT

Practice for lesson 1.3

Business problem: Create a small tasks API.

Steps

1. Create a Postgres database and a tasks table.
2. Add an Express app with GET and POST /tasks.
3. Validate input before saving.
4. Test with Postman.

Expected output: Creating a task returns 201; listing returns the saved tasks.

SOLUTION (ANSWER KEY)

Use a parameterised query or the ORM client so user input never goes into SQL text. Return 400 with a message when the title is empty.

Check yourself

1. Why use parameterised queries?
   Answer: They keep user input from being run as SQL, which prevents SQL injection.
2. What is middleware?
   Answer: A function that runs before the route handler, such as auth or validation.

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